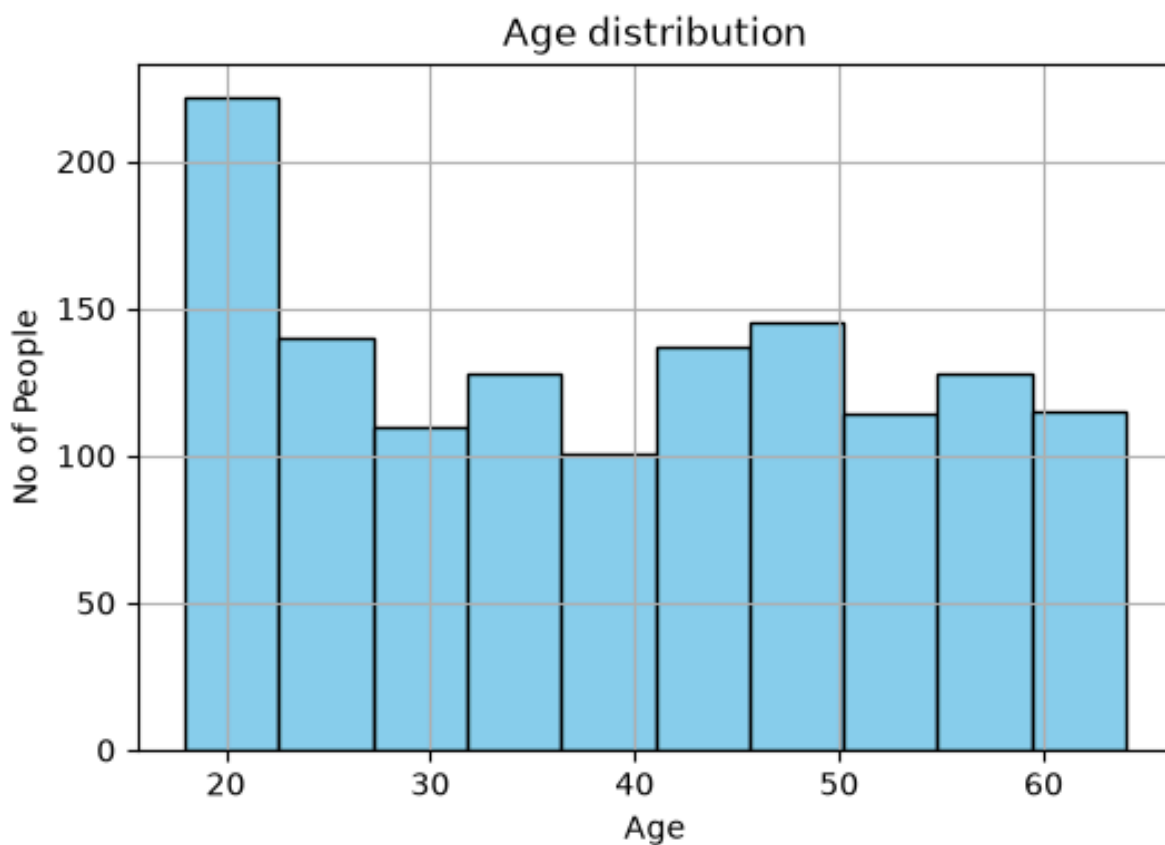


```

1  import pandas as pd
2  import seaborn as sns
3  import numpy as np
4  import matplotlib.pyplot as plt
5
6
7  from sklearn.datasets import load_wine
8
9  df = pd.read_excel('insurance.xlsx')
10 df.head()
11
12 wine = load_wine()
13 print(wine)
14 df = pd.DataFrame(wine.data, columns = wine.feature_name)
15
16 df['target'] = wine.target
17 df
18
19
20 #histogram plot plotting No of Age
21
22 plt.figure(figsize=(6,4))
23 plt.hist(df['age'], bins=10 , color='skyblue', edgecolor = 'black')
24
25 plt.title('Age distribution')
26 plt.xlabel('Age')
27 plt.ylabel('No of People')
28
29 plt.grid(True)
30 plt.show()
31

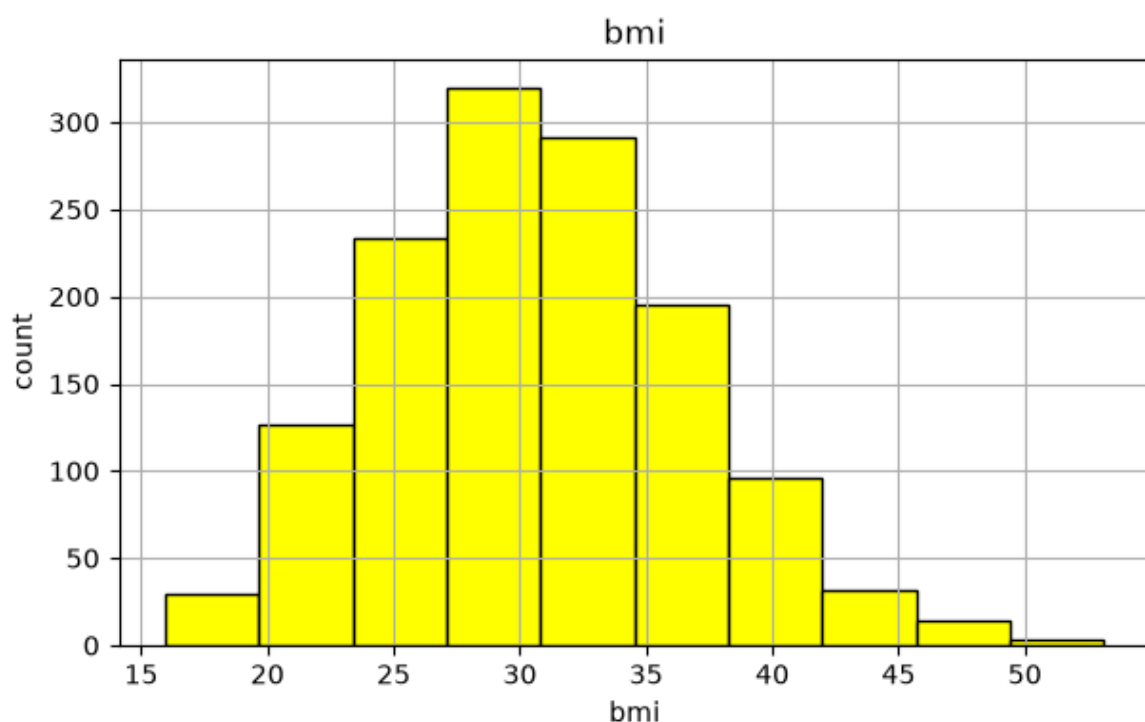
```

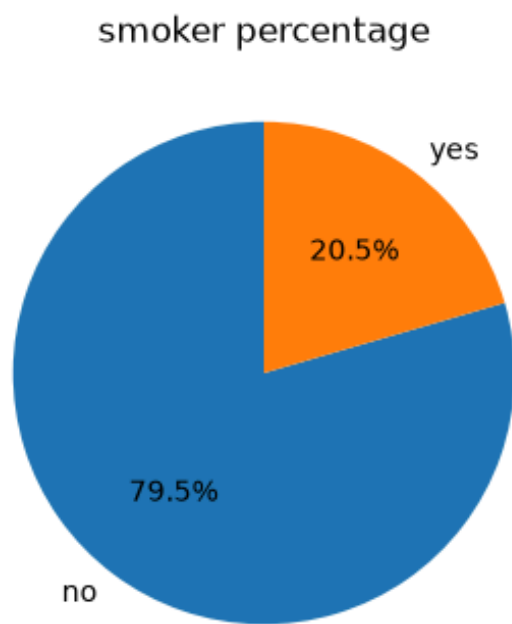
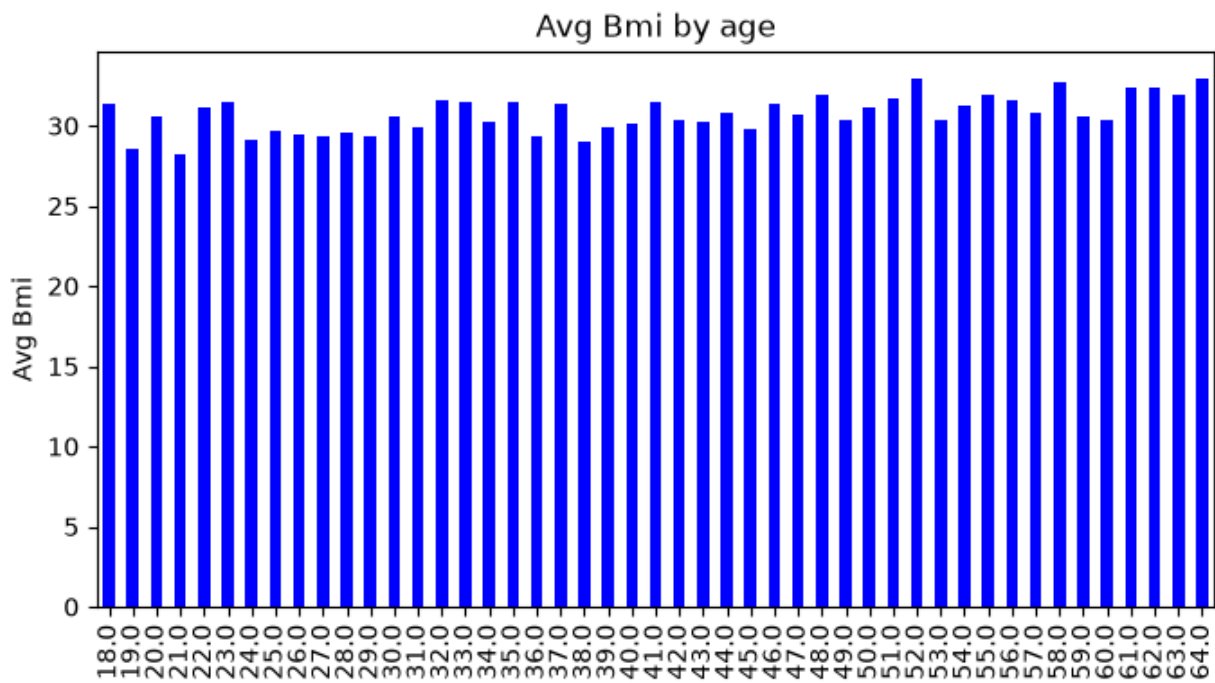


```

33 #histogram plot plotting bmi count
34 plt.figure(figsize=(7,4))
35 plt.hist(df['bmi'],bins=10,color='yellow', edgecolor='black')
36
37 plt.title('bmi')
38 plt.xlabel('bmi')
39 plt.ylabel('count')
40
41 plt.grid(True)
42 plt.show()
43
44
45
46 #bar chart groupby age Vs their avg bmi
47 plt.figure(figsize=(8,4))
48 df.groupby('age')['bmi'].mean().plot(kind='bar',color='blue')
49
50 plt.title('Avg Bmi by age')
51 plt.xlabel('Age')
52 plt.ylabel('Avg Bmi')
53 plt.show()
54
55
56
57 #Pie chart counting smoker and non-smokers
58 plt.figure(figsize=(8,4))
59 df['smoker'].value_counts().plot(kind='pie',autopct='%1.1f%%',startangle=90)
60 plt.title('smoker percentage')
61 plt.ylabel('')
62 plt.show()
63
64

```





```

63
64
65 #corelation heatmap
66 numeric_df = df.select_dtypes(include='number')
67 plt.figure(figsize=(8,7))
68 sns.heatmap(numeric_df.corr(),annot=True,cmap='coolwarm',fmt='.2f')
69 plt.title('correlation heatmap')
70 plt.show()
71
72
73 # Box plot Charges by Smoker
74 plt.figure(figsize=(8,7))
75 sns.boxplot(x='smoker',y='charges', data=df)
76 plt.xlabel('smoker')
77 plt.ylabel('charges')
78 plt.show()
79
80
81 #Scatter plot by Bmi by sex
82 plt.figure(figsize=(6,4))
83 sns.scatterplot(x='age',y='bmi', hue='sex', data=df)
84 plt.title('scatter')
85 plt.show()

```

